Notebook Functionalities

A plain-language guide to the Jupyter notebooks in the Data_Science project

This guide covers 18 notebooks found in the repository. It explains each notebook's main job and how the notebooks relate to the project workflow.

How to use this guide

Start with data loading and preparation when you need to build or refresh datasets. Use the exploration notebooks to understand the data. Use the model and comparison notebooks after the required daily data table is ready. Some notebooks are alternative workflows and do not all need to be run in sequence.

Main data paths

> The project includes both older all_regions workflows and newer daily-table workflows. Several modeling notebooks read the aqi_daily MySQL table. The hourly-to-daily conversion notebook builds and updates the daily pollutant-and-weather data. Check each notebook's setup cells for the exact input table or file before running it.

1. Data loading and preparation

air-quality-classification/notebooks/DB_Connection.ipynb

Connects to MySQL, finds and previews station CSV files, and imports station data into database tables. It is an initial data-loading notebook.

air-quality-classification/notebooks/etl_data_processor.ipynb

Combines station tables, checks missing values and duplicates, cleans the combined readings, loads the result into the all_regions table, and runs a few SQL summaries.

air-quality-classification/notebooks/clean_readings.ipynb

Checks a CSV for record identity problems, duplicates, and invalid measurements, then writes a validated cleaned CSV. This is a file-based cleaning workflow.

air-quality-classification/notebooks/hourly_to_daily_pollutant_conversion.ipynb

Cleans hourly readings, aggregates them into daily station readings, joins daily ERA5 weather, appends the newer daily pollutant-and-weather data through 2025, and saves the combined data to CSV and MySQL.

CNEMC_Figshare_2014-2025/beijing_combined_daily_2014-2025.ipynb

Converts the newer Beijing hourly pollutant readings to daily readings and aligns their columns with the older daily pollutant file. This creates and reviews pollutant-only daily CSVs.

CNEMC_Figshare_2014-2025/make_beijing_daily_pollutant_weather_dataset.ipynb

Converts the hourly pollutant data to daily station rows, joins the daily ERA5 weather features, and saves both the pollutant-only and pollutant-plus-weather CSV files.

air-quality-classification/notebooks/aqi_labeling_and_splits.ipynb

Calculates daily China AQI values and six AQI categories from pollutant readings, examines the effect of interpolated values, and prepares chronological training and test sets. It can optionally save those sets to MySQL.

2. Data exploration and statistics

air-quality-classification/notebooks/eda_beijing_air_quality.ipynb

Explores pollutant and weather measurements with charts, including averages over time and comparisons across Beijing monitoring stations.

air-quality-classification/notebooks/stats_analysis_beijing_air_quality.ipynb

Summarizes pollution levels by station, checks guideline exceedances, and uses statistical tests, including ANOVA and follow-up comparisons, to examine station differences.

3. AQI prediction and evaluation

air-quality-classification/notebooks/air_quality_model.ipynb

Builds a next-day AQI category model from the daily pollutant-and-weather database table. It calculates labels and targets, tunes and evaluates XGBoost against a baseline, demonstrates prediction, and saves a model artifact and dataset.

2 class/two_class_aqi.ipynb

Trains and evaluates a two-class next-day model: clean versus polluted. It reads daily data from MySQL, uses a chronological holdout, and saves model and evaluation files.

3 class/three_class_aqi.ipynb

Trains and evaluates a three-class next-day AQI model: clean, light-to-moderate pollution, and heavy pollution. It reads daily data from MySQL and saves metrics, reports, and a model artifact.

air-quality-classification/notebooks/comprehensive_model_evaluation.ipynb

Evaluates two-, three-, and six-class next-day predictions on a shared chronological holdout. It reports class-level metrics and confusion matrices, and summarizes what the results mean.

4. Comparing candidate model families

air-quality-classification/notebooks/two_class_model_comparison.ipynb

Compares several model families for clean-versus-polluted next-day prediction, along with simple baselines. It reports class, ranking, and probability-calibration metrics and gives a final recommendation.

air-quality-classification/notebooks/three_class_model_comparison.ipynb

Compares candidate model families for three-class next-day AQI prediction. It includes baselines, per-class reports, confusion matrices, and a heavy-pollution alert review.

air-quality-classification/notebooks/six_class_model_comparison.ipynb

Compares candidate model families for all six next-day AQI categories. It includes baselines, per-class results, confusion matrices, and a severe-pollution alert review.

5. Weather-model analysis

air-quality-classification/notebooks/weather_feature_walkthrough.ipynb

Builds daily examples from hourly pollutant data, adds weather summaries, compares pollutant-only and weather-enabled models, checks performance across time periods, stations, and seasons, and can train the weather-enabled app artifact.

weather_feature_walkthrough.ipynb (repository root)

A separate saved copy of the weather walkthrough. It covers the same broad weather-feature workflow, but it is not identical to the notebook under air-quality-classification/notebooks.